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Construction and Validation of the Reminiscence Functions Scale

Jeffrey Dean Webster

Department of Psychology, Vancouver Community College, Langara Campus, British Columbia.

This research introduces the Reminiscence Functions Scale (RFS), a 43-item questionnaire that can be used to assess reminiscence functions over the lifecourse. Adults (710) ranging in age from 17 to 91 (mean age = 45.76 years) completed a 54-item Reminiscence Functions Scale-prototype measure, the results of which were submitted to a principal components analysis. Results indicated the viability of a 43-item, 7-factor solution with good reliability. Factors were labeled: Boredom Reduction, Death Preparation, Identity/Problem-Solving, Conversation, Intimacy Maintenance, Bitterness Revival, and Teach/Inform. A separate validity study demonstrated the predictive validity of the RFS. Directions for future research are discussed.

THE process of recapturing salient memories from one's personal past continues to interest researchers concerned with naturalistic memory dynamics (Webster & Cappeliez, 1993). As a vehicle for examining ecologically valid memory, reminiscence offers researchers a window into not only the mechanics of memory, but its functions as well. Unfortunately, although myriad potentially adaptive functions of reminiscence have been postulated in the literature (e.g., improving self-esteem, coping with depression, self-understanding, death preparation, etc.), there currently exists no reliable and valid questionnaire measure of multiple reminiscence functions.

Despite the rapid expansion of published articles concerning some aspect of reminiscence, many reviewers (Haight, 1991; Kovach, 1990; Merriam, 1980; Molinari & Reichlin, 1984–1985; Moody, 1988; Romaniuk, 1981; Thornton & Brotchie, 1987; Webster & Cappeliez, 1993) have indicated that there remain several important obstacles to future progress. These faults concern problems with operational definitions, sampling characteristics, and methodological shortcomings. This article addresses the latter concern, specifically the lack of psychometrically sound instrumentation.

Lack of Valid and Reliable Measures

One of the most common stumbling blocks to integration of existing evidence is the paucity of psychometrically sound instruments for assessing various reminiscence dimensions. Without such assessment tools, researchers have frequently been forced to assemble ad hoc questions tailored to their own unique research agendas. In the majority of cases, reliability and validity data have either not been presented, or are of questionable quality. In order for meaningful comparisons between research projects to occur, clinical and experimental investigators need a readily available, psychometrically sound instrument that produces comparable scores across studies.

To my knowledge, there has been only one published attempt to construct a paper-and-pencil measure of reminiscence functions. Romaniuk and Romaniuk (1981) developed a measure called the Reminiscence Uses Scale (RUS), a 13-item questionnaire in which subjects responded either yes, no, or not sure to the uses of reminiscence listed in the scale. According to Romaniuk and Romaniuk, all items were intuitively generated by the researchers following suggestions contained in the literature. The final sample in the study was composed of 91 highly educated older adults (mean age = 78.5, range = 58–98) who were ~80% female and were living in one of three new retirement communities. A factor analysis revealed three factors — labeled (1) Self-Regard/Image Enhancement, (2) Present Problem-Solving, and (3) Existential/Self-Understanding — accounting for 17.7%, 16.6%, and 13.2% of the total variance, respectively.

Romaniuk and Romaniuk's (1981) work represents a good initial attempt to identify reminiscence uses in a homogeneous sample of elderly adults. It suffers from several limitations, however, which render it less than optimal for its stated purpose. The most serious shortcomings are listed below. In terms of its sample, the preponderance of females precludes an analysis on the basis of gender; the ethnic composition is not given, eliminating a discussion of potential ethnic differences; and the homogeneous nature of the sample in terms of education, health, and age characteristics limits its generalizability.

In terms of its psychometric properties, the coefficients of internal consistency are modest (i.e., .64, .69, and .57 for factors 1, 2, and 3, respectively), the total number of items constituting the scale are truncated at 13, which limits the number of factors that could potentially emerge, and some inclusion criteria for item retention are somewhat problematic. For example, several factor loadings are quite conservative (i.e., below .49), and items were assigned to factors on the basis, "that the factors make intuitive or theoretical sense" (p. 480). The latter is cause for some consternation, because item 5 in the scale (i.e., to entertain) loaded .48 on factor 3 and .47 on factor 1, yet was included, without adequate explanation, in the latter factor. Finally, no validity checks using external, psychometrically established measures in a predictive or concurrent fashion were conducted.

Recently, Wong and Watt (1991) discussed how various types of reminiscence may be differentially related to successful aging. Through content coding of verbatim reminiscence protocols, they derived six types of reminiscence that they labeled: integrative, instrumental, transmissive, escapist, obsessive, and narrative. Their results suggest that higher levels of integrative and instrumental, and lower levels of obsessive, reminiscence types are related to successful aging.

Wong and Watt's classification system is a welcome shift toward methodological refinement and an increased focus on the multidimensional nature of reminiscence procedures. Unfortunately, there are two disadvantages to their approach. First, it is relatively time-consuming and labor-intensive, because protocols must be elicited from subjects in a face-to-face format, and raters must be trained to code the resulting qualitative data.

Second, their work introduces a crucial confound between a taxonomic division, or *type*, and a specific memorial use, or *function*. In the operational definitions of their types, which they explicitly state are "mutually exclusive categories" (p. 276), they include a host of functions that overlap taxonomic divisions. For instance, the function of increasing self-esteem is a characteristic of three types (i.e., integrative, transmissive, and escapist) and life satisfaction enhancement is a core component of both integrative and instrumental types.

To illustrate the potential danger of operational overlap discussed above, consider a specific coding example provided by the authors. One subject reflected on how the marital breakup of his or her parents caused resentment as a teenager, but that now the subject had achieved reconciliation with them due to the subject's more mature outlook. This passage is coded as integrative because it "conveys the idea of resolving past conflicts" (p. 276). This seems reasonable enough, but it does not demonstrate exclusivity, because another function of such an excerpt might be to pass on a moral lesson (e.g., do not judge others too harshly without full understanding). If the latter is the case, then the above scenario is clearly an example of transmissive reminiscence, because moral instruction and personal wisdom are fundamental constituents of this type. Despite this concern, Wong and Watt (1991) provide interrater reliability data (i.e., kappa = .88 for two judges) that would seem to attenuate this criticism.

Finally, Wong and Watt's (1991) taxonomy is derived from elderly adults all older than 65 (mean age = 77.8 years), once again potentially limiting the generalizability of the results to the aged.

In summary, there has been only one previous attempt at constructing a paper-and-pencil measure of reminiscence functions. Perhaps due to the limitations identified above, it has been rarely used in published studies of reminiscence. The purpose of the present article is to develop a psychometrically sound instrument that improves on earlier work.

METHOD

Development of the Reminiscence Functions Scale-Prototype (RFS-p)

Pilot study 1: item generation. — As part of earlier work (Webster, 1989) and classroom demonstrations, participants were asked to write down two different reasons why they reminisced, and two different reasons why other people might reminisce. From this larger pool of respondents, 40 subjects (19 males and 21 females) were selected. The respondents ranged in age from 18 to 76 years (mean age = 42.42, SD = 18.84), with approximately equal numbers falling in each age category (i.e., Young: N = 12, range = 18–29; Middle-age: N = 14, range = 30–49; Old: N = 14, range = 50–76). The sample had a mean education level of 12.55 years (range = 8–17, SD = 1.78) and a mean self-perceived health rating of 5.27 (range = 3–7, SD = 1.17), where scores could range from 1 = "very poor" to 7 = "excellent." In terms of marital status, 15%, 30%, 37.5%, and 17.5% of the sample were separated, single, married, and divorced, respectively. The sample was predominantly Caucasian (77.5%), followed by Chinese (10.0%), Native Indian (5.0%), East Indian (2.5%), and Other (2.5%). Between them, these subjects generated a total of 115 statements to the stems: "I reminisce because. . . ," and "Others reminisce because. . . ." The verbatim statements were then edited for spelling and to make them grammatically flow from the stem, "I reminisce: . . . ." Subsequently, all 115 statements were readministered to a new group of subjects.

Pilot study 2: item reduction/selection. — A new group of 119 subjects selected from an introductory psychology class at Vancouver Community College, a demographically diverse educational institution, were asked to rate each of the 115 statements, generated in Pilot Study 1, on a 6-point scale where 1 = "never reminisce" (for the stated reason) to 6 = "very frequently reminisce" (for the stated reason). The sample included 44 males and 72 females ranging in age from 18 to 46 years (mean age = 22.62 years, SD = 4.84) with a mean education level of 13.05 years (range = 8–17, SD = 1.26). In terms of ethnic composition, 41.18%, 39.5%, 1.68%, 0.84%, 5.04%, and 11.76% were Caucasian, Chinese, Japanese, Black, East Indian, and Other, respectively.

Two means of culling redundant items were used in this phase of instrument construction. First, an informal principal components factor analysis was performed on the 115 statements. The number of factors was set a priori at 10 in order to both ensure that as wide a net as possible was cast at this early stage of instrument development, while at the same time limiting the number of potential factors to a theoretically and pragmatically workable size. Items were then ranked within each factor in terms of their loading.

Second, a research assistant was instructed to hand-sort each statement into between 5–10 groupings, subsequently producing eight relatively distinct clusters. Comparing the overlap between the two methods resulted in a rationally derived compromise of nine groupings of six items each. If a factor had more than six items, the top six with the highest loadings from the informal factor analysis were included. If a factor had less than six items, all the original items were included and additional items were written that seemed to tap the same underlying dimension. At this point, the groupings were given the tentative labels of: Boredom Reduction, Death Preparation, Identity Consolidation, Problem-Solving, Conversation, Intimacy Maintenance, Obsessive/Pathological, Self-Esteem Enhancement, and Teach/Inform.

In summary, a two-stage process of item generation and reduction/selection yielded a prototype instrument with the following strengths: the items were generated by a diverse sample spanning age, sex, education, ethnic, and other demographic dimensions; it covered an expansive range of potential functions; items reflected input from subjects themselves as well as input directed by theoretical considerations indicated in the literature; and it had high face validity. The next step involved administration of the finalized prototype to a new sample.

Measures

The Reminiscence Functions Scale prototype (RFS-p) is a 54-item questionnaire in which subjects are asked to respond on a 6-point scale how often they reminisce with a particular function in mind. Subjects read the following introduction:

> At different points throughout their lives, most adults think about their past. Recalling earlier times can happen spontaneously or deliberately, privately or with other people, and may involve remembering both happy and sad episodes. The process of recalling memories from our personal past is called reminiscence, an activity engaged in by adults of all ages.
>
> This questionnaire concerns the why, or functions, of reminiscence. That is, what purpose does reminiscence fulfil, or, what goal does retrieving certain memories help you accomplish?

The 54 items were randomly ordered in the questionnaire and presented as completions to the stem: "When I reminisce it is: . . .". For example, an item following the stem might read: "to pass the time during idle or restless hours." Responses ranged from 1 = "never" (reminisce for the stated purpose) to 6 = "very frequently" (reminisce for the stated purpose). Subjects recorded their answers on an attached answer sheet by filling in the circle containing the number that best represented their choice (i.e., 1–6). The questionnaire typically took between 15–25 min to complete.

Procedure

Subjects were recruited through a variety of means, including hand delivery of questionnaires to retirement housing complexes and community centers; volunteers from introductory psychology classes at Vancouver Community College and St. Thomas University; and having students solicit volunteers from the ranks of family members, peers, and neighbors. The bulk of respondents were gained from the latter category, where students received nominal class credit for soliciting volunteers from the community. All subjects volunteered, and it was stressed to students that if potential subjects they were attempting to solicit appeared reluctant in any way, even after it had been emphasized that responses were anonymous and that there was no obligation to complete any or all of the questionnaire, that students would thank the person for their time and select an alternate person. Information concerning the voluntary and anonymous nature of the questionnaire was explicitly detailed in a cover letter that contained two telephone numbers that persons could contact about concerns or comments. Only questionnaires with all 54 RFS-p questions answered completely were used. In this manner, 710 usable questionnaires were obtained.

Subjects

Subjects consisted of a diverse convenience sample, including 289 males and 421 females ranging in age from 17 to 91 years (mean age = 45.76, *SD* = 21.69). In terms of education, the sample ranged from those with no formal schooling whatsoever, to those with at least some postgraduate work (mean education = 12.57, *SD* = 2.58). Self-perceived health scores ranged from 1 = "very poor" to 7 = "excellent" (mean health = 5.09, *SD* = 1.27). Table 1 provides a breakdown for the above demographic variables for the sample as a whole and decomposed by age decade.

RESULTS

Exploratory Principal Components Analysis

The RFS-p was submitted to a principal components analysis with varimax rotation using Systat for the PC (version 5.1; Wilkinson, 1990). The criteria adopted for item and factor retention were that: (a) eigenvalues must have a minimum value of 1, (b) minimal item loading on a factor was set at ≥ .50, and (c) at least 4 or more items meeting the ≥ .50 criterion would be the minimum to constitute a factor (in order to facilitate reliability checks). Using these criteria and rotating the loadings to facilitate interpretation produced seven clearly distinct factors, labeled as follows: (1) Boredom Reduction, (2) Death Preparation, (3) Identity/Problem-Solving, (4) Conversation, (5) Intimacy Maintenance, (6) Bitterness Revival, and (7) Teach/Inform. A total of 43 of the original 54 items were retained in the analysis. Table 2 lists the factor loadings on each factor for all 43 RFS questions.

The seven factors that emerged closely mirror those produced by the informal analysis during the RFS-p development, with two major exceptions being (1) the elimination of a self-esteem factor and (2) the collapsing together of the Identity Consolidation and Problem-Solving groupings. With respect to exception (1), three items (i.e., Q18: remembering previous achievements makes me feel proud of myself; Q34: reexperience feelings of pride due to past accomplishments; and Q44: to build up my self-esteem by recalling earlier life successes) had respective factor loadings of .76,

Table 1. Demographic Characteristics for Total Sample and By Decade

Sample Demographics	Age (by decade) 10s	20s	30s	40s	50s	60s	70s	80s[a]	Total
Gender									
Male	31	32	39	38	41	44	44	20	289
Female	88	73	47	39	39	58	58	18	421
Age									
Mean	18.3	22.5	34.2	44.4	53.8	64.6	73.1	83.2	45.7
SD	.6	2.6	2.7	2.6	2.7	2.8	2.6	2.8	21.6
Education									
Mean	12.5	13.1	13.5	13.4	13.4	12.0	11.3	10.3	12.6
SD	.6	1.0	1.9	2.4	2.5	3.4	3.7	2.9	2.6
Health									
Mean	5.3	5.3	5.2	4.9	5.2	5.1	4.8	4.9	5.1
SD	1.1	1.2	1.1	1.3	.9	1.3	1.6	1.7	1.3

[a]One 91-year-old male was included in the 80s decade.

.72, and .61, qualifying them on two of the inclusion criteria outlined earlier. However, their intercorrelations were low (range = .40–.60), and reliability estimates for only three items would be suspect. They were, therefore, eliminated from further analysis, although they were clearly suggestive of a self-esteem/pride factor that warrants further investigation. Abbreviated RFS item examples for all seven RFS factors are given in Table 3. The complete RFS can be obtained by writing to the author.

Table 2. Reminiscence Functions Scale Factor Loadings

	Factor Loadings						
RFS Items	1	2	3	4	5	6	7
Factor 1							
Q20	.83	.05	.05	.07	.05	.19	−.01
Q47	.80	.16	.04	.14	.03	.08	−.04
Q14	.73	.09	.09	.09	.07	.16	−.12
Q26	.73	.12	.10	.07	.01	.20	.11
Q4	.66	.08	.18	.10	.13	.11	−.03
Q24	.61	.04	.26	.10	.07	.20	−.01
Factor 2							
Q41	.16	.76	.17	.06	.11	.07	.07
Q43	.09	.76	.07	.09	.15	.00	.24
Q37	.10	.74	.12	.04	.13	.06	.19
Q48	.27	.65	.26	.09	.06	.14	.03
Q2	.02	.59	.12	.10	.12	.00	.21
Q12	−.00	.58	.11	.07	.13	.06	.27
Factor 3							
Q40	.11	.14	.73	−.06	−.01	.15	−.04
Q49	.10	.18	.73	.17	.08	−.01	.09
Q39	.06	.18	.70	.15	.10	.07	.07
Q45	.02	.14	.64	.06	−.09	.13	.05
Q23	−.02	.01	.64	.22	.12	.06	.09
Q15	.27	.04	.64	.09	.11	.15	−.08
Q52	.10	.03	.63	.10	.01	.27	.01
Q33	.03	.17	.59	.02	.13	.11	.18
Q5	.12	.09	.56	.10	−.06	−.02	−.03
Q11	.09	−.13	.53	.11	−.09	.19	.02
Q13	.05	.20	.52	.01	.04	.12	.10
Q31	.12	.20	.50	.13	.06	−.00	.20
Factor 4							
Q42	.19	.10	.13	.71	.04	.07	.17
Q28	.06	.04	.19	.68	.11	−.01	.21
Q7	.09	.16	.17	.67	.07	.07	−.01
Q36	.18	−.00	.12	.66	−.00	.06	.26
Q9	.06	.17	.20	.61	.13	−.01	.03
Factor 5							
Q6	.04	.16	.04	.10	.84	−.00	.16
Q32	.05	.16	.01	.05	.83	.04	.23
Q17	.17	.26	.02	.11	.74	.12	.14
Q51	.05	.04	.14	.04	.59	.21	.06
Factor 6							
Q53	.18	.16	.11	−.01	.07	.82	−.10
Q50	.21	.12	.11	.06	.04	.80	−.06
Q16	.18	.09	.14	.02	.20	.68	−.07
Q22	.20	−.10	.25	−.00	.01	.60	.14
Q19	.23	−.01	.19	.12	−.02	.59	.02
Factor 7							
Q29	−.05	.25	.06	.13	.12	.01	.78
Q1	−.00	.28	−.08	.02	.17	−.03	.72
Q35	−.00	.24	.10	.22	.14	−.04	.63
Q25	−.00	−.02	.37	.21	.07	.01	.59
Q38	−.08	.38	.04	.08	.25	−.12	.58

Reliability

As a means of assessing factor reliability, internal consistency scores were computed using coefficient alpha for all items. Alpha levels and factor intercorrelations for the RFS are reported in Table 4.

As can be seen, the internal consistency of the factors is good, ranging from .79 for Conversation to .89 for Identity/Problem-Solving (mean alpha = .84, *SD* = .04). The

Table 3. Sample Items From All 7 Factors of the Reminiscence Functions Scale

Factor 1:	Boredom Reduction
Q20	to reduce boredom (.83)[a]
Q47	for something to do (.80)
Factor 2:	Death Preparation
Q41	because I feel less fearful of death after I finish reminiscing (.77)
Q43	because it helps me see that I've lived a full life and can therefore accept death more calmly (.76)
Factor 3:	Identity/Problem-Solving
Q40	to try to understand myself better (.73)
Q49	to see how my strengths can help me solve a current problem (.73)
Factor 4:	Conversation
Q42	to create ease of conversation (.71)
Q28	to create a common bond between old and new friends (.68)
Factor 5:	Intimacy Maintenance
Q6	to keep alive the memory of a dead loved one (.84)
Q32	to remember someone who has passed away (.83)
Factor 6:	Bitterness Revival
Q53	to keep memories of old hurts fresh in my mind (.82)
Q50	to rekindle bitter memories (.80)
Factor 7:	Teach/Inform
Q29	in order to teach younger persons about cultural values (.78)
Q1	to teach younger family members what life was like when I was young and living in a different time (.72)

[a]Values in parentheses are factor loadings.

Table 4. Factor Intercorrelation Matrix and Internal Consistency Data for the Reminiscence Functions Scale

	Factor						
Factor	1	2	3	4	5	6	7
1	—	.32	.36	.35	.24	.48	.05*
2		—	.39	.34	.40	.23	.52
3			—	.44	.23	.40	.28
4				—	.29	.20	.41
5					—	.22	.42
6						—	.02*
7							—
	Internal Consistency						
Alpha	.88	.85	.89	.79	.84	.82	.82

Note. Except where indicated, all correlations are significant at the $p <$ .0001 level.

$^*p$ = not significant.

intercorrelations indicate that all but two factors on the RFS are significantly correlated with one another, ranging from small to moderate in magnitude (*M* correlation = .31, *SD* = .13, range = .02–.52). On average, the variance accounted for between factors is < 10%, indicating an expected association between factors that is nevertheless small enough to validate the divergent nature of the subscales.

Predictive Validity

Two separate means of assessing the predictive validity of the RFS were conducted. The first method examined the relationships between reminiscence functions and personality traits. The second investigated developmental differences in reminiscence functions over the life span.

Personality. — A new and separate sample of 123 introductory psychology students at Vancouver Community College served as subjects. Participants included 45 males and 78 females ranging in age from 18 to 45 (mean age = 24.6, *SD* = 6.9). Although there is a dearth of empirical studies examining the relationship between personality characteristics and reminiscence dimensions (Fry, 1992; Webster & Cappiliez, 1993), what limited evidence does exist suggests there may be a relatively stable relationship between various dimensions of personality and reminiscence. For instance, both Fry (1992) and Webster (in press) found that the personality domain of openness was associated with simple reminiscence frequency.

For the validity study, Costa and McCrae's (1985) NEO-FFI (Form S) personality measure was used as the dependent variable in a series of regression analyses using the RFS factors as predictor variables. The NEO-FFI was chosen because it is an extensively used instrument of high reliability and validity, normed on a sample spanning the entire adult age spectrum, and because it lends itself to specific hypotheses vis-à-vis reminiscing.

The NEO-FFI measures five personality domains, three of which have been extensively researched and used in the RFS validity study. The Neuroticism (N) domain comprises the six facets of anxiety, hostility, depression, self-consciousness, impulsiveness, and vulnerability. Persons scoring high on this domain are more likely to manifest high levels of anxiety and negative emotions such as anger and embarrassment. The Extraversion (E) domain comprises the six facets of warmth, gregariousness, assertiveness, activity, excitement-seeking, and positive emotions. Persons scoring high on this domain are more likely to manifest high levels of sociability, assertiveness, and stimulation-seeking. The Openness (O) domain comprises the six facets of fantasy, aesthetics, feelings, actions, ideas, and values. Persons scoring high on this domain are more likely to manifest high levels of introspection, exploration of novelty, and intellectual curiosity.

Given the constituents of the three personality domains, the following a priori hypotheses were tested: namely, that (a) Factor 6 would be positively correlated with Neuroticism, (b) Factor 4 would be positively correlated with Extraversion, and (c) Factor 3 would be positively correlated with Openness.

The results support these predictions, providing preliminary evidence of the predictive validity of the RFS. Specifically, Factor 6 correlated .42 with Neuroticism ($p < .01$, $n = 120$, one-tailed), Factor 4 correlated .33 with Extraversion ($p < .01$, $n = 123$, one-tailed), and Factor 3 correlated .18 with Openness ($p < .05$, $n = 120$, one-tailed).

Developmental. — Age differences on the RFS, full details of which can be found in Webster's current work (available on request), also provide evidence of its predictive validity. Because some authors (Kiernat, 1983; Webster & Young, 1988) explicitly call for a life-span approach to reminiscence research, it would be instructive to examine possible age differences in the RFS. Two specific examples will suffice here. First, there is an age difference in Factor 2 (Death Preparation). Consistent with Butler's (1963) notion of the life review, one of the functions of reminiscence in advanced old age is to prepare to accept one's impending biological demise and, therefore, it is expected that elderly adults, in general, being closer to death than their younger counterparts, would use reminiscence for this reason more frequently. We can see from Figure 1 that the results are clear and strongly support this prediction. Specifically, there is a monotonic increase in the use of reminiscence in the service of death preparation with age ($F = 18.24$, df = 7, 693, $p < .0001$). Second, there is also a significant relationship ($F = 26.19$, df = 7, 693, $p < .0001$) between age and Factor 7 (Teach/Inform). We can see from Figure 2 that older adults use reminiscence in this fashion with greater frequency than younger adults. This finding is consistent with the greater accumulation of experience as a corollary of advanced chronological age. In other words, older adults' extensive navigation through multiple life events has pro-

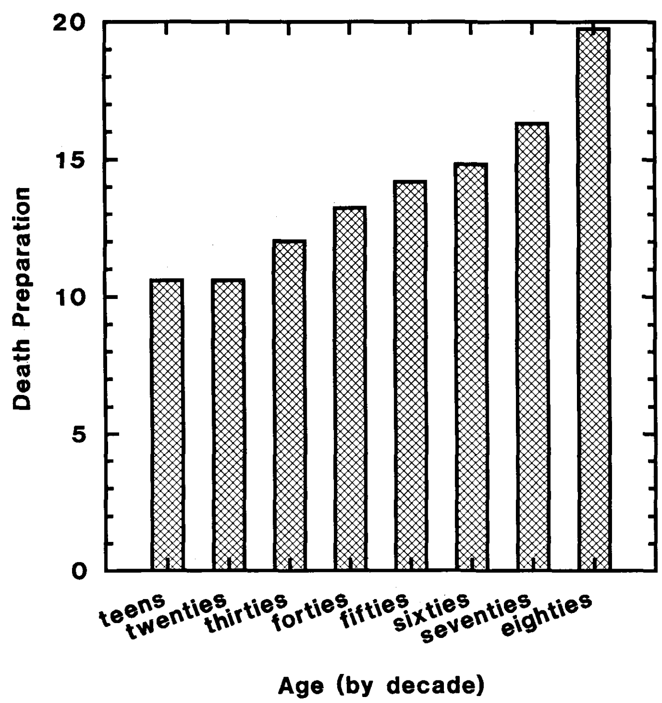

Figure 1. Age differences on the RFS Death Preparation subscale.

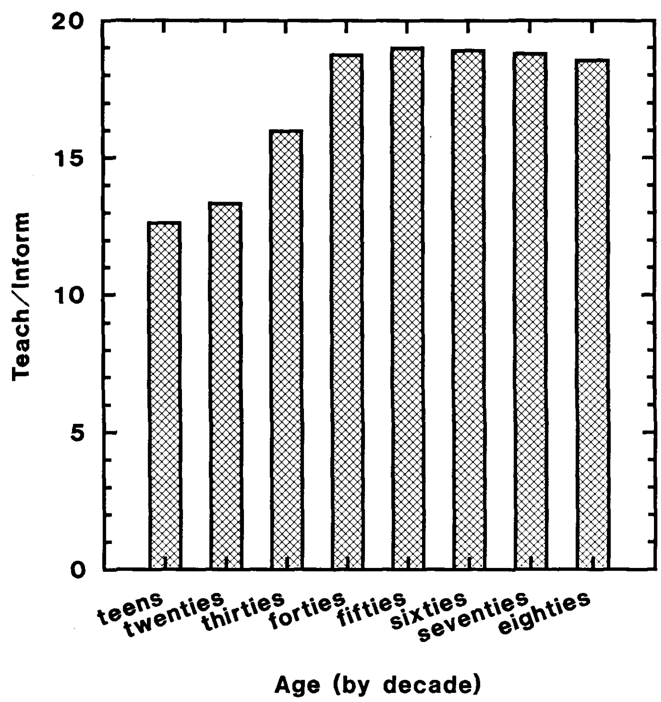

Figure 2. Age differences on the RFS Teach/Inform subscale.

vided a wealth of knowledge on which they may draw. The relatively sharp break from youth to middle adulthood may be a consequence of parenthood more than age per se.

DISCUSSION

The RFS is a valid, reliable, and relatively comprehensive measure of many fundamental uses of reminiscence in adulthood. It builds on, strengthens, and elaborates earlier work by Romaniuk and Romaniuk (1981), and is consistent with many findings of Wong and Watt's (1991) recent taxonomy.

There exist clear parallels between the present findings and Romaniuk and Romaniuk's (1981) earlier RUS. Both measures derive a Problem-Solving component and an Identity or Existential/Self-Understanding factor. In Romaniuk and Romaniuk's scale, these emerged as separate factors, whereas they were combined into a single factor in the RFS. This combination in the RFS is somewhat problematic and needs further refinement for the following reasons.

First, Identity and Problem-Solving items were assumed to constitute separate subscales during the second pilot study and were expected to remain distinct. Second, an inspection of how the items clustered within the single scale revealed that they are not randomly intermixed, but rather tended to group together with problem-solving items generally grouping together first, followed by the cluster of identity items. Third, other factor rotation patterns (i.e., equimax) produce distinct factors for Identity and Problem-Solving. Finally, the Romaniuks' work demonstrates that these two functions may indeed constitute separate subscales.

These concerns suggest that future refinements of the RFS may attempt to tease apart the Identity/Problem-Solving factor into separate subscales, allowing for finer grained analysis of certain hypotheses. As only one example, problem-solving has been postulated to be higher in middle-age than in old age (Lieberman & Falk, 1971). Such a finding may be masked in the RFS if there is an opposite effect for Identity, whereby a higher Identity score among elderly adults would cancel out the lower problem-solving score.

Further, consistent with the Romaniuks' RUS, a self-esteem factor emerged in the RFS, but, as noted above, did not meet all the inclusion criteria established for factor retention. In future refinement of the RFS, additional items reflecting self-esteem components might be included in an attempt to meet the inclusion criteria established previously. Alternatively, the inclusion criteria themselves could be less stringent, but this would seem to defeat the major aim of this research, which is to develop an assessment instrument whose psychometric properties are as high as possible.

Parallels also exist between the RFS and Wong and Watt's (1991) typology. The RFS has several advantages over the latter typology, including ease of administration and scoring, greater comprehensiveness in terms of identified factors, and clearer differentiation between factors. Nevertheless, findings of the present study are clearly consistent with many facets of Wong and Watt (1991). Specifically, strong similarities emerge between (1) Factor 6 (Bitterness Revival) of the RFS and the Obsessive type, (2) Factor 7 (Teach) and the Transmissive type, and (3) The Integrative and Instrumental types which, in combination, match closely with Factor 3 (Identity/Problem-Solving). Less strongly related are Factor 1 (Boredom Reduction) and the Escapist type. The latter involves a clear deprecation of the present relative to the past, whereas the former probably involves more pedestrian recollections of a positive nature. Finally, inasmuch as one of the main functions of Narrative reminiscence is to "recount past anecdotes that may be of interest to the listener" (p. 274), it shares certain characteristics with Factor 4 (Conversation). Factor 2 (Death Preparation) and Factor 5 (Intimacy Maintenance) find no counterparts in Wong and Watt's (1991) typology, although one suspects that there should be a correlation between Factor 2 of the RFS and the Integrative type. The fact that two separate methods should result in such similar findings regarding reminiscence functions bodes well for the convergent validity of the RFS.

Many of the resulting factors of the RFS have at one point been nominated as potentially important uses of reminiscence in elderly adults (e.g., Factor 2: Butler, 1963; Factor 3: Lieberman & Tobin, 1983; Factor 4: Boden & Bielby, 1983). The RFS allows researchers to assess multiple functions simultaneously and allows for direct comparisons across studies.

Future research directions can include contrasting reminiscence functions between groups of interest (e.g., on the basis of sex, age, culture, religion, marital status, living arrangements, health, personality, etc.). As a new instrument, the RFS will find its initial place as a descriptive, research-based tool. Eventually, however, it may find employment as a diagnostic measure as well. For instance, is there an RFS "profile" consistent with clinical depression, or perhaps predictive of the early stages of dementing illnesses such as Alzheimer's disease? It is hoped that these and other important questions can be fruitfully explored, in part, with the RFS.

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Address correspondence and reprint requests to Jeffrey Dean Webster, Department of Psychology, Vancouver Community College, Langara Campus, 100 West 49th Avenue, Vancouver, British Columbia, Canada V5Y 2Z6.

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